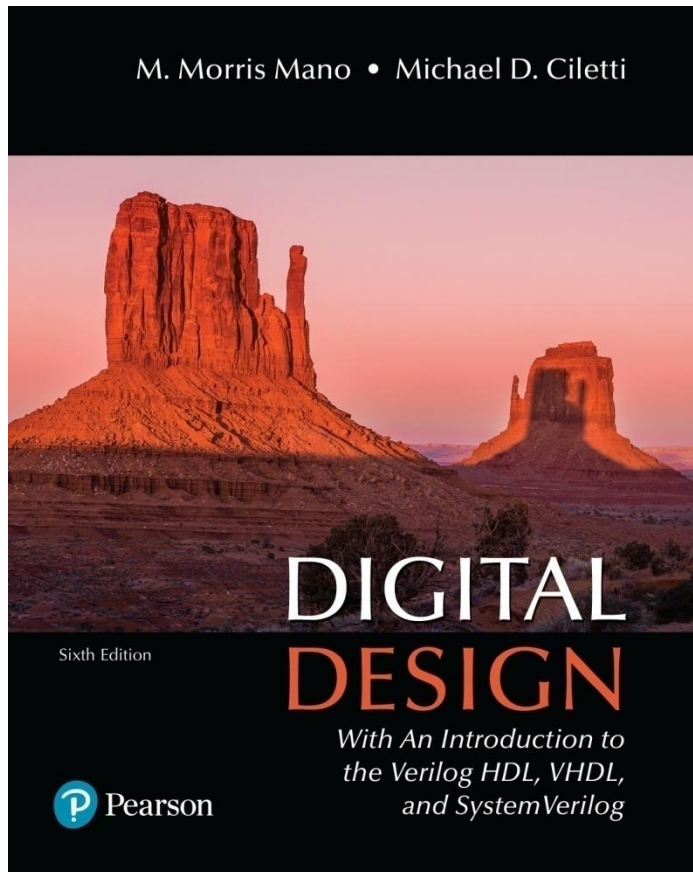


Digital Design

With an Introduction to the Verilog HDL, VHDL, and SystemVerilog

6th Edition



Chapter 09

Laboratory Experiments with Standard ICs and FPGAs

Figure 9.1
Digital gates in IC packages with identification numbers and pin assignments.

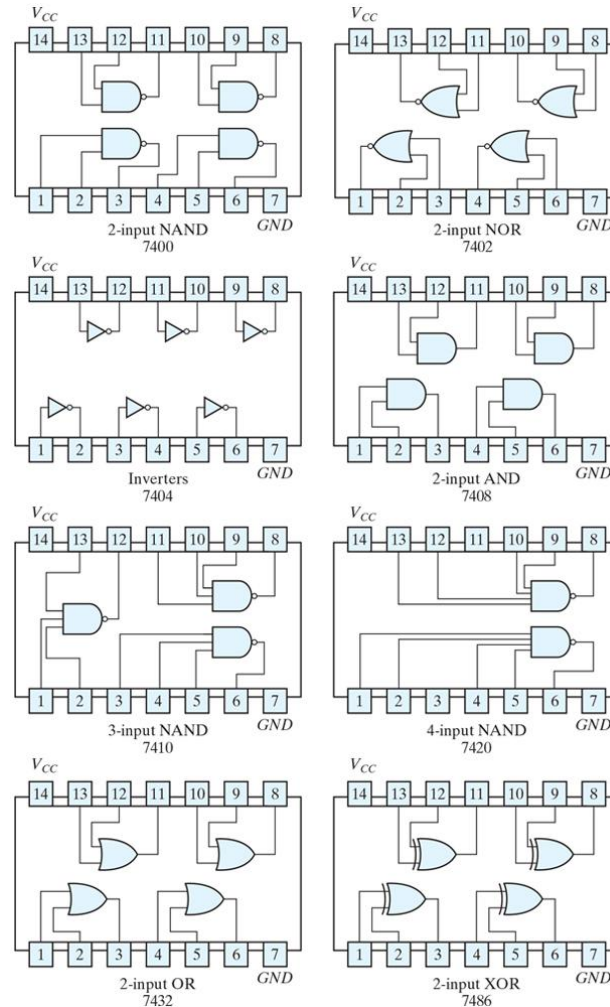


Figure 9.2

IC type 7493 ripple counter.

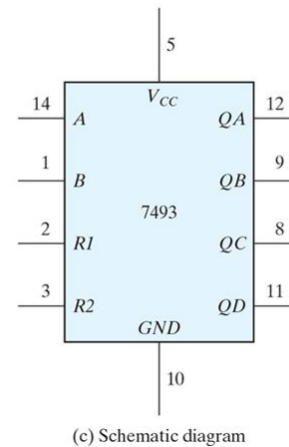
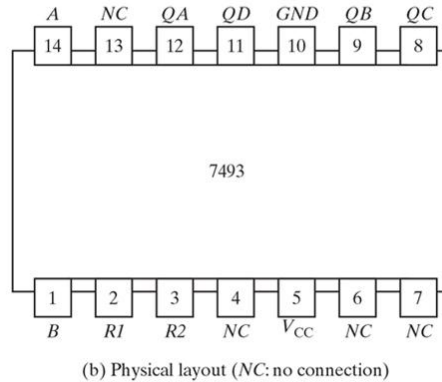
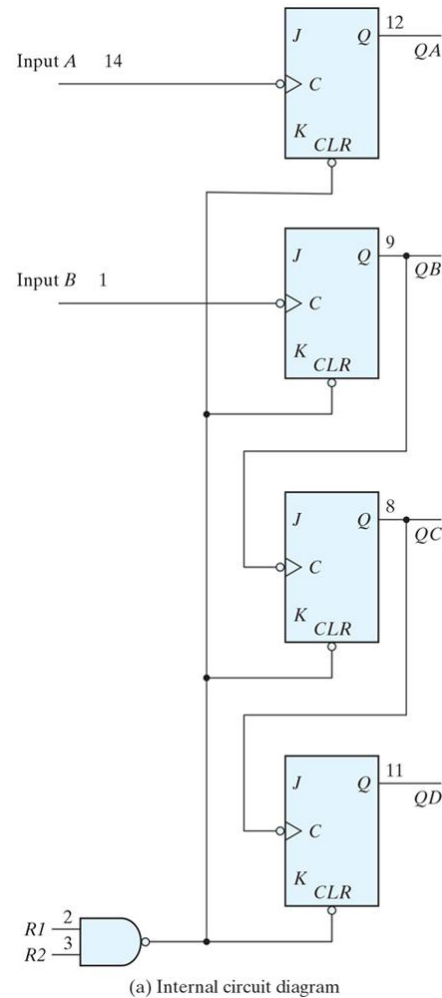


Table 9.1

Integrated Circuits Required for the Experiments.

IC Number	Description	Graphic Symbol	
		In Chapter 9	In Chapter 10
	Various gates	Fig. 9.1	Fig. 10.1
7447	BCD-to-seven-segment decoder	Fig. 9.8	—
7474	Dual <i>D</i> -type flip-flops	Fig. 9.13	Fig. 10.9(b)
7476	Dual <i>JK</i> -type flip-flops	Fig. 9.12	Fig. 10.9(a)
7483	Four-bit binary adder	Fig. 9.10	Fig. 10.2
7493	Four-bit ripple counter	Fig. 9.2	Fig. 10.13
74151	8 × 1 multiplexer	Fig. 9.9	Fig. 10.7(a)
74155	3 × 8 decoder	Fig. 9.7	Fig. 10.6
74157	Quadruple 2 × 1 multiplexers	Fig. 9.17	Fig. 10.7(b)
74161	Four-bit synchronous counter	Fig. 9.15	Fig. 10.14
74189	16 × 4 random-access memory	Fig. 9.18	Fig. 10.15
74194	Bidirectional shift register	Fig. 9.19	Fig. 10.12
74195	Four-bit shift register	Fig. 9.16	Fig. 10.11
7730	Seven-segment LED display	Fig. 9.8	—
72555	Timer (same as 555)	Fig. 9.21	—

Figure 9.3
Binary counter.

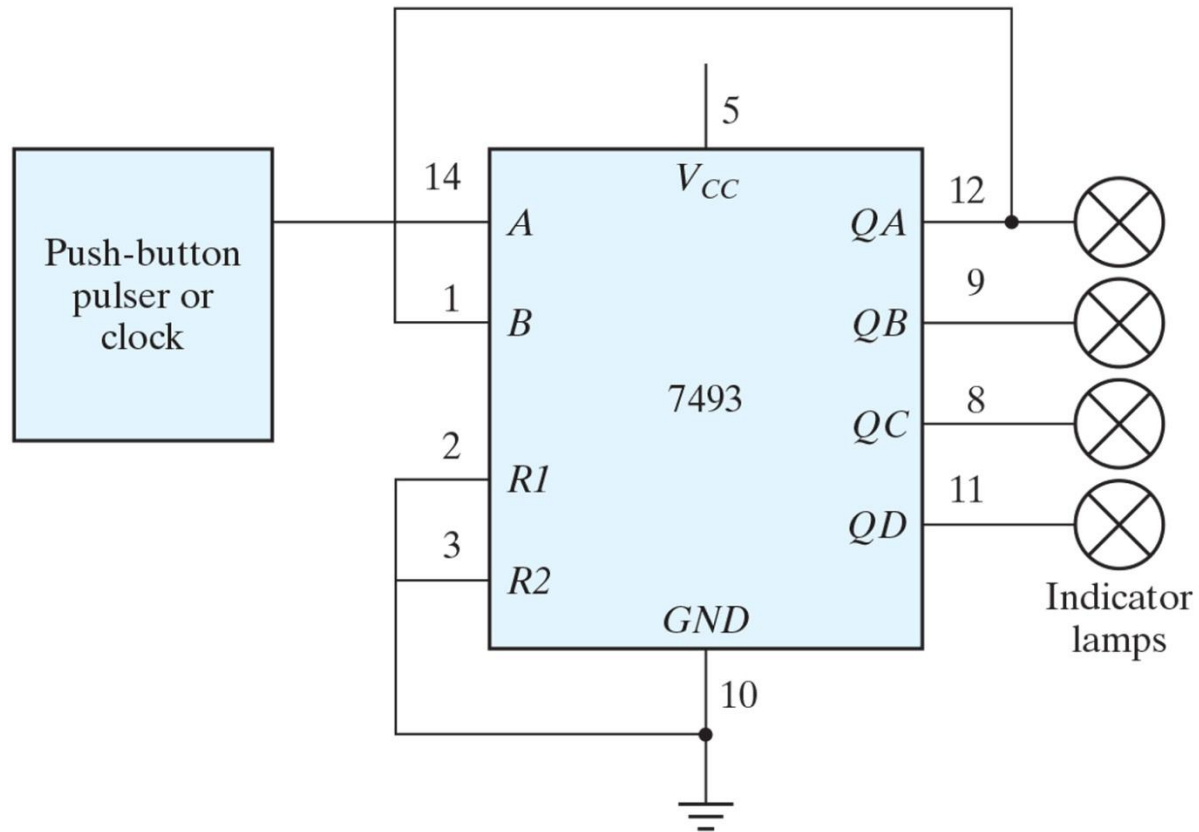


Figure 9.4
BCD counter.

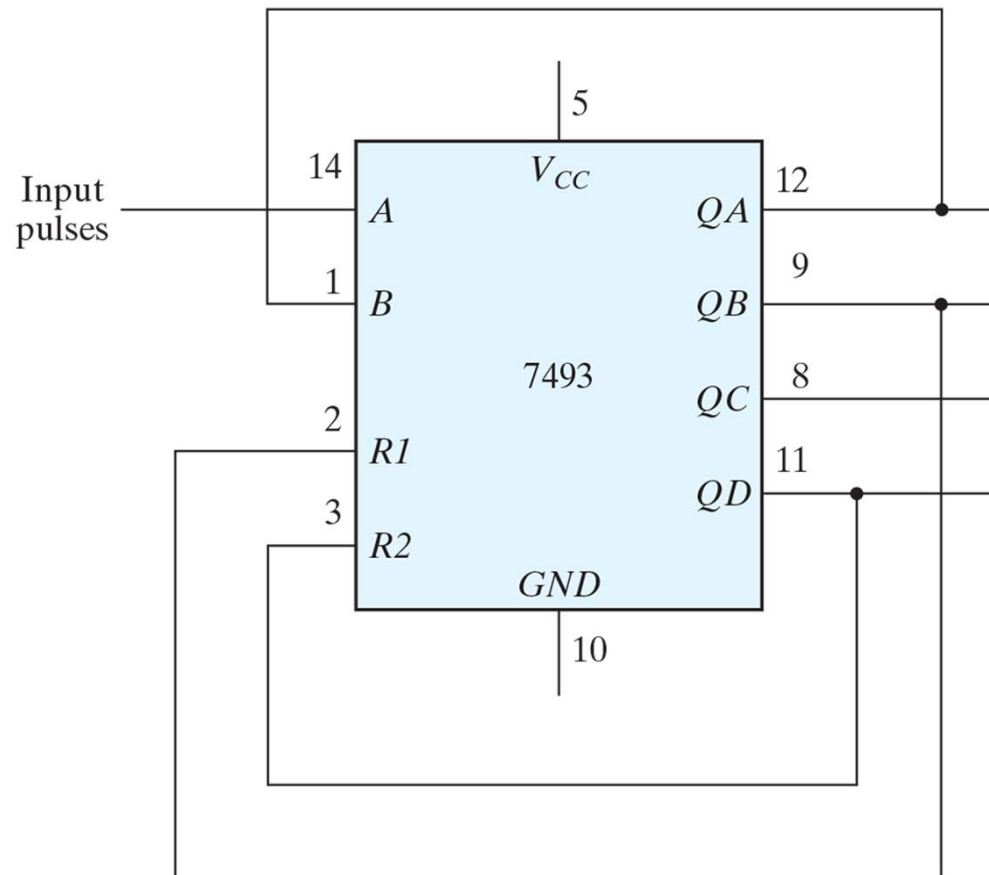


Figure 9.5
Waveforms for NAND gate.

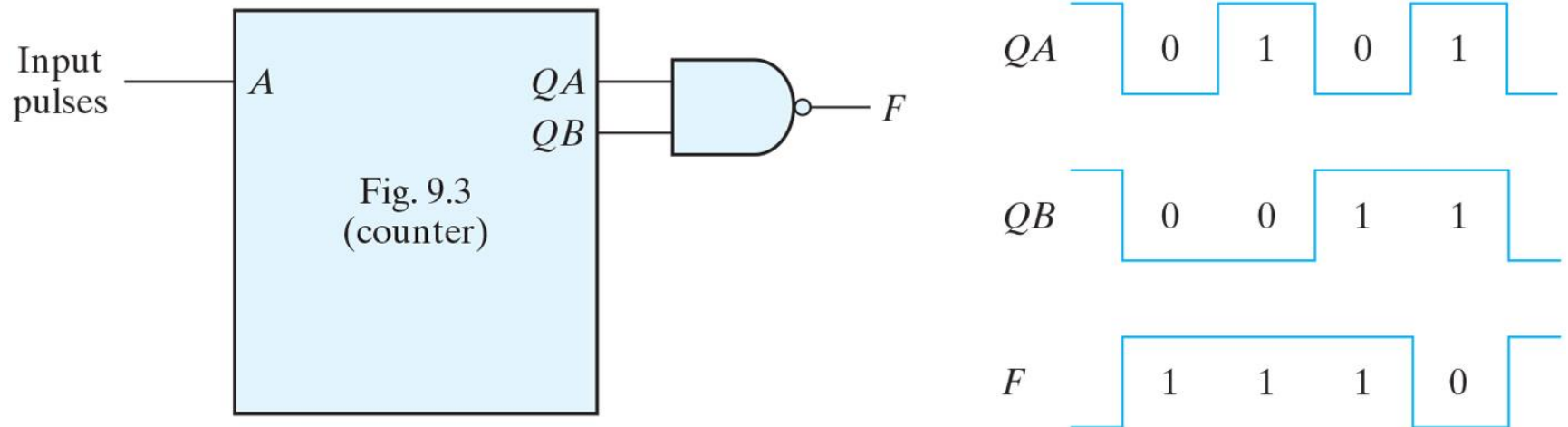


Figure 9.6
Logic diagram for Experiment 3.

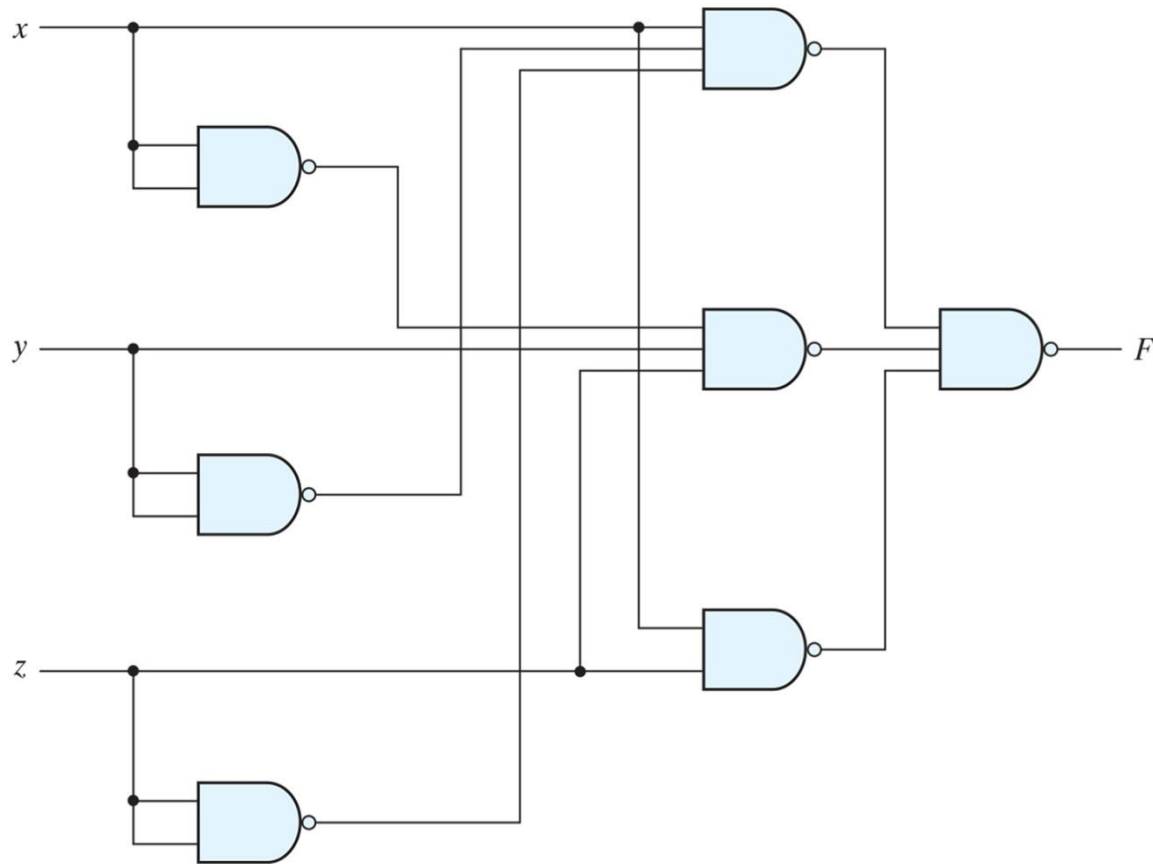
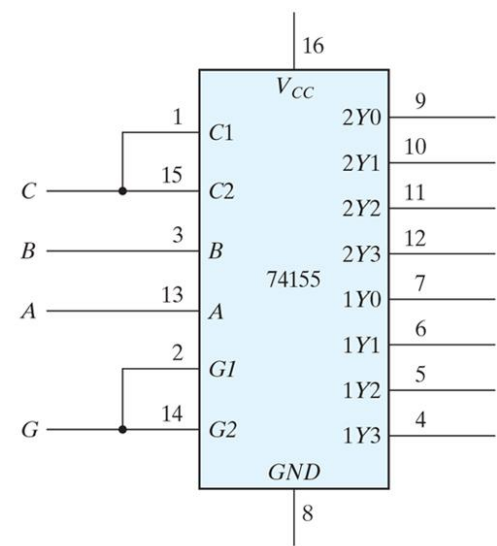


Figure 9.7

IC type 74155 connected as a 3 × 8 decoder.



Truth table												
Inputs				Outputs								
G	C	B	A	2Y0	2Y1	2Y2	2Y3	1Y0	1Y1	1Y2	1Y3	
1	X	X	X	1	1	1	1	1	1	1	1	1
0	0	0	0	0	1	1	1	1	1	1	1	1
0	0	0	1	1	0	1	1	1	1	1	1	1
0	0	1	0	1	1	0	1	1	1	1	1	1
0	0	1	1	1	1	1	0	1	1	1	1	1
0	1	0	0	1	1	1	1	0	1	1	1	1
0	1	0	1	1	1	1	1	1	0	1	1	1
0	1	1	0	1	1	1	1	1	1	0	1	1
0	1	1	1	1	1	1	1	1	1	1	0	1

Figure 9.8
BCD-to-seven-segment decoder (7447) and seven-segment display (7730).

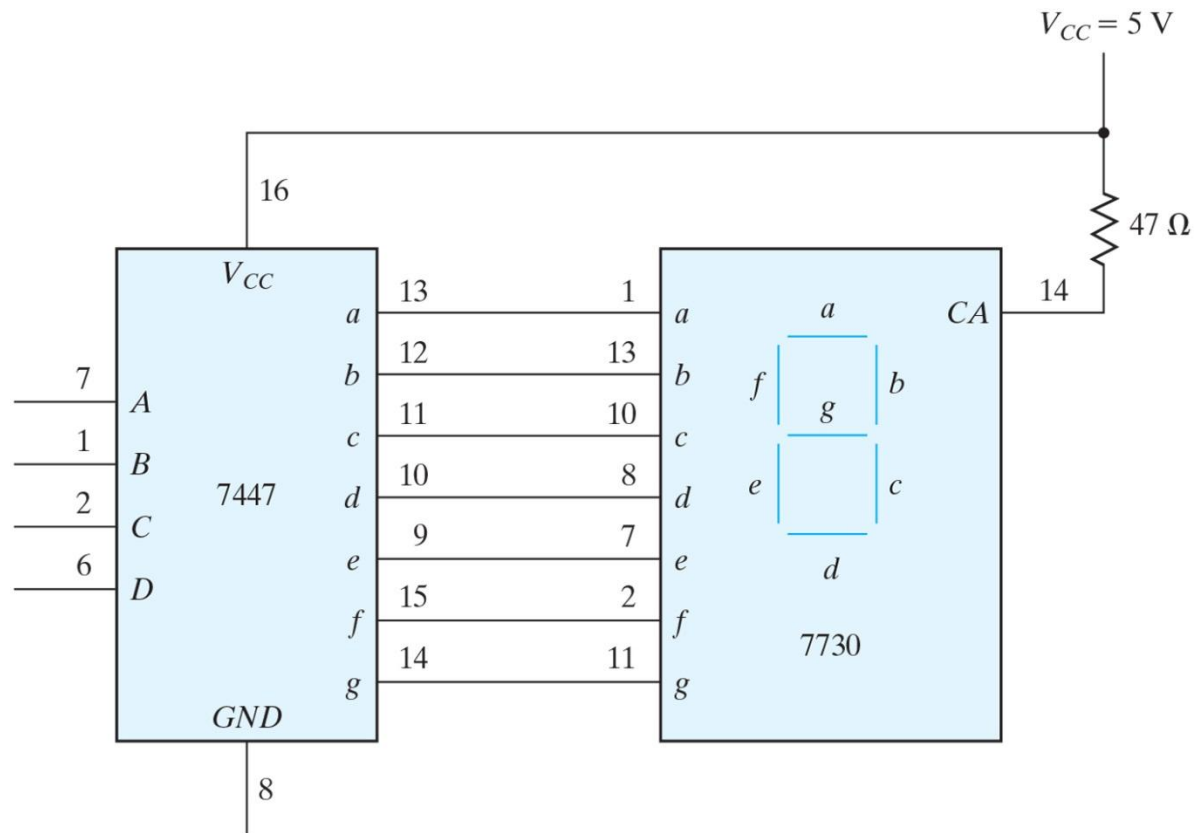
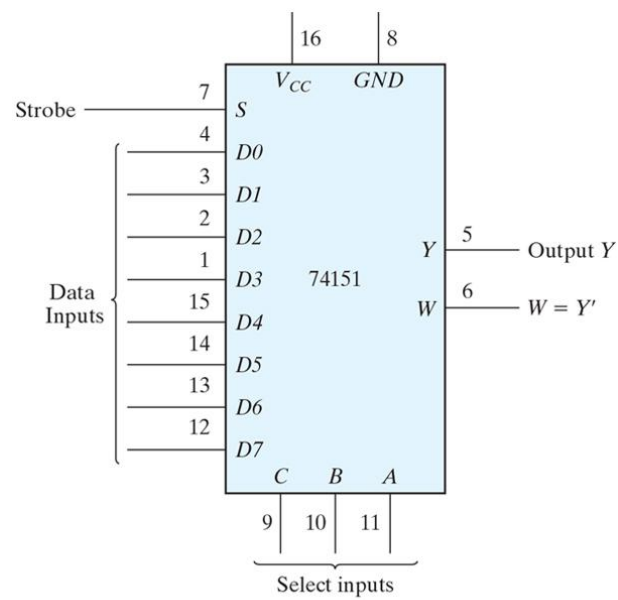


Figure 9.9
IC type 74151 8 × 1 multiplexer.



Function table				
Strobe S	Select			Output Y
	C	B	A	
1	X	X	X	0
0	0	0	0	$D0$
0	0	0	1	$D1$
0	0	1	0	$D2$
0	0	1	1	$D3$
0	1	0	0	$D4$
0	1	0	1	$D5$
0	1	1	0	$D6$
0	1	1	1	$D7$

Figure 9.10
IC type 7483 four-bit binary adder.

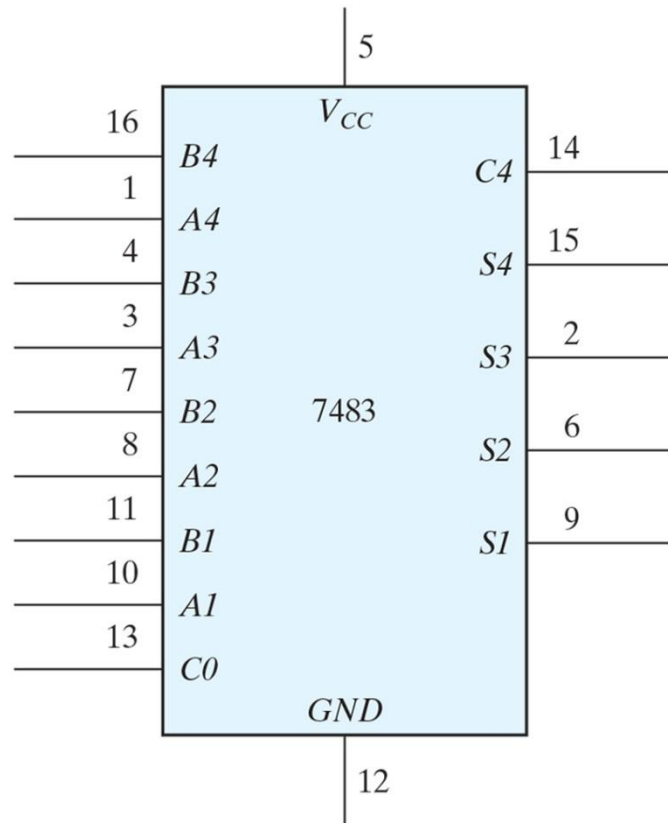


Figure 9.11

Four-bit adder–subtractor.

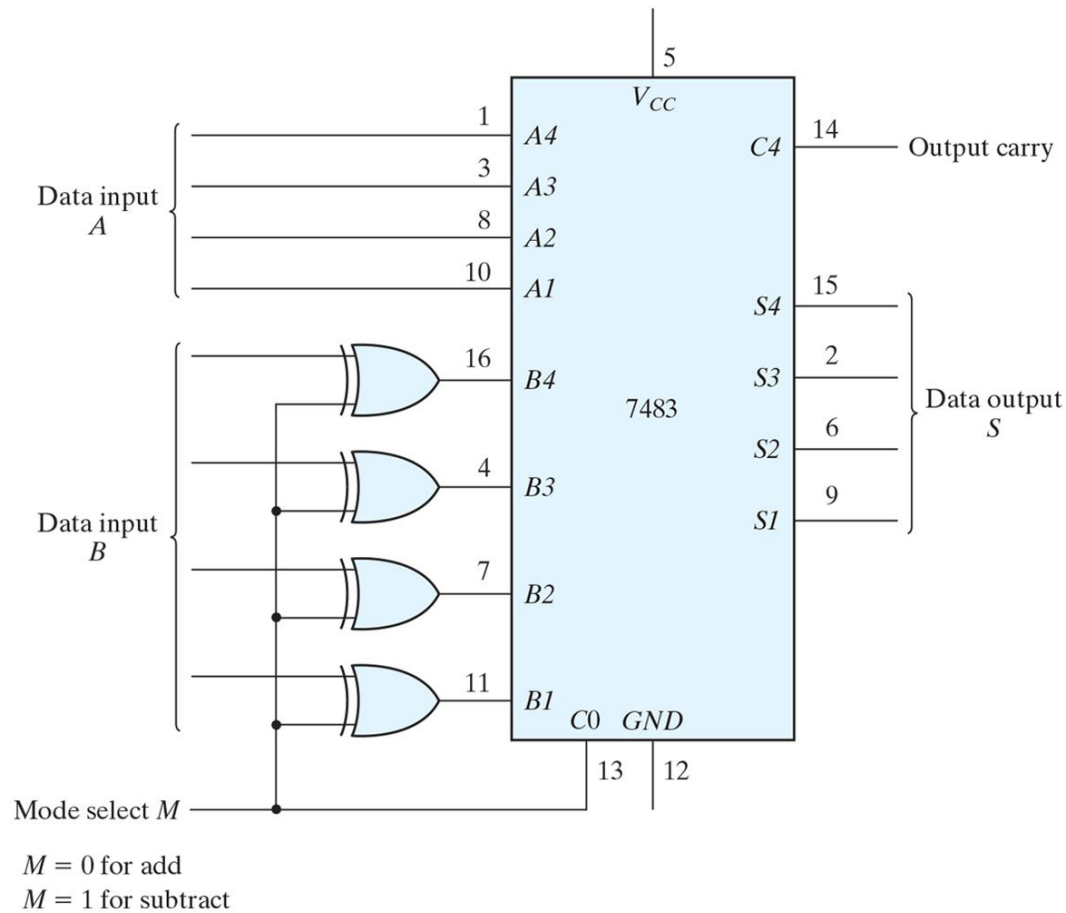
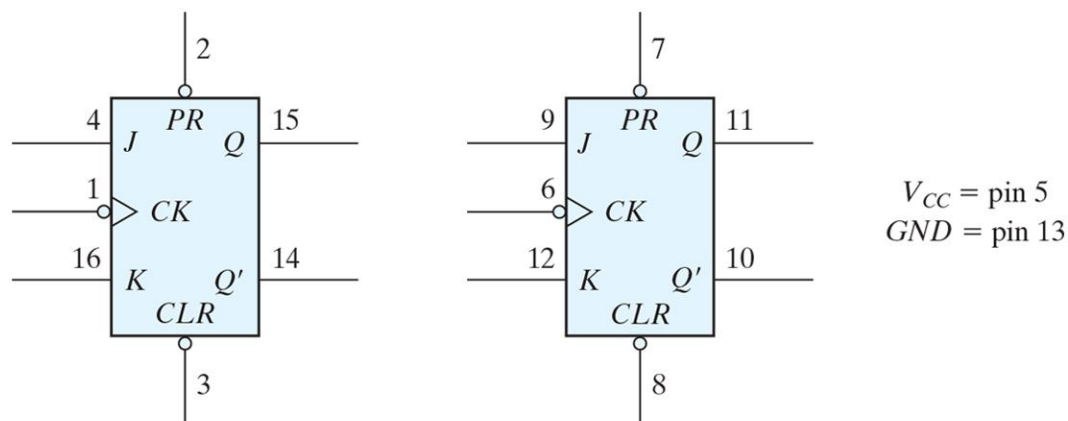


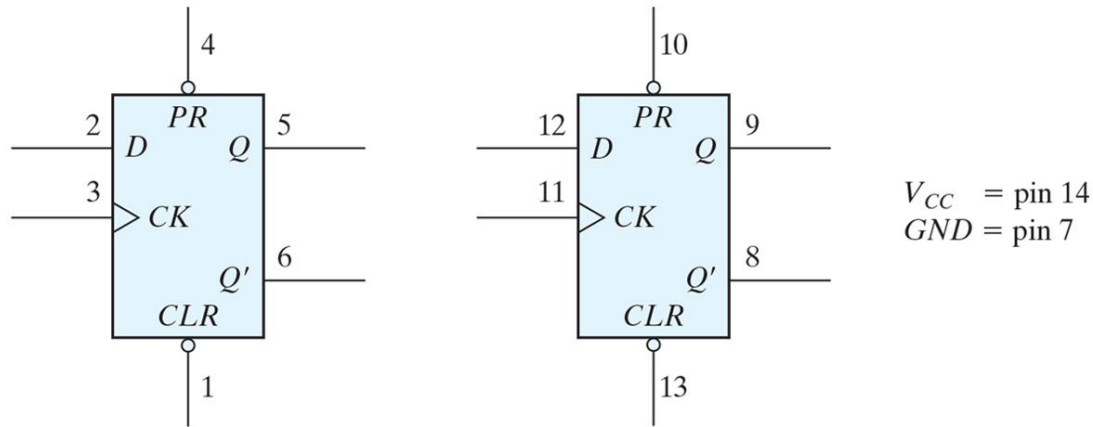
Figure 9.12
IC type 7476 dual *JK* master–slave flip-flops.



Function table

Inputs					Outputs	
Preset	Clear	Clock	<i>J</i>	<i>K</i>	<i>Q</i>	<i>Q'</i>
0	1	X	X	X	1	0
1	0	X	X	X	0	1
0	0	X	X	X	1	1
1	1		0	0	No change	
1	1		0	1		
1	1		1	0	1	0
1	1		1	1	Toggle	

Figure 9.13
IC type 7474 dual *D* positive-edge-triggered flip-flops.



Function table

Inputs				Outputs	
Preset	Clear	Clock	<i>D</i>	<i>Q</i>	<i>Q'</i>
0	1	X	X	1	0
1	0	X	X	0	1
0	0	X	X	1	1
1	1	↑	0	0	1
1	1	↑	1	1	0
1	1	0	X	No change	

Figure 9.14
State diagram for Experiment 9.

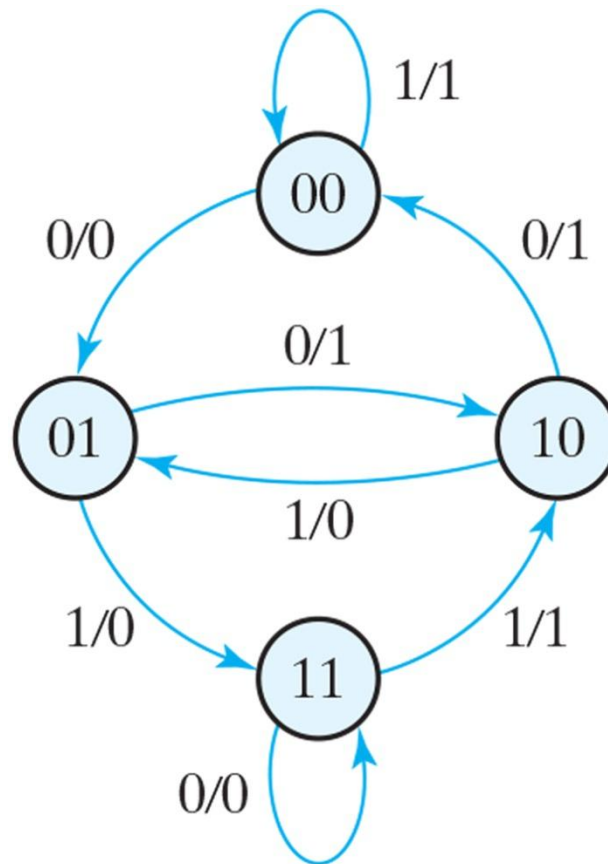
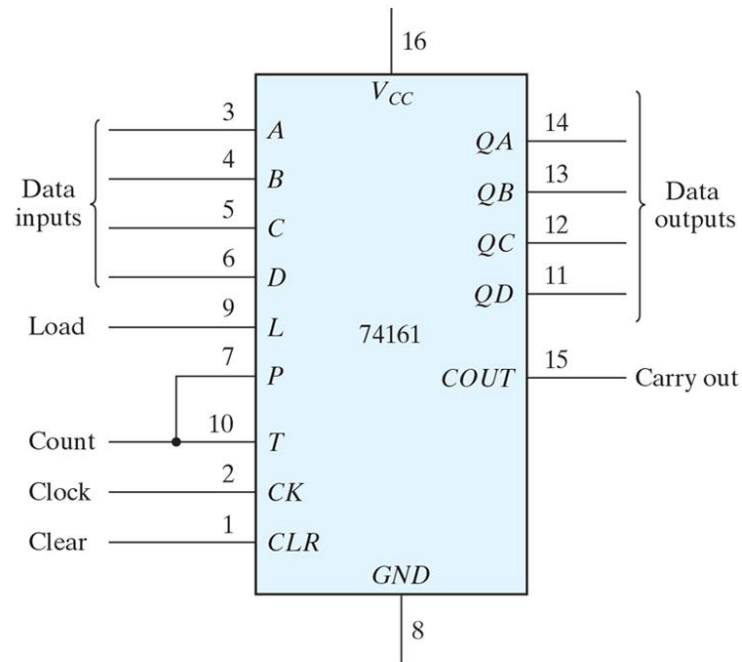


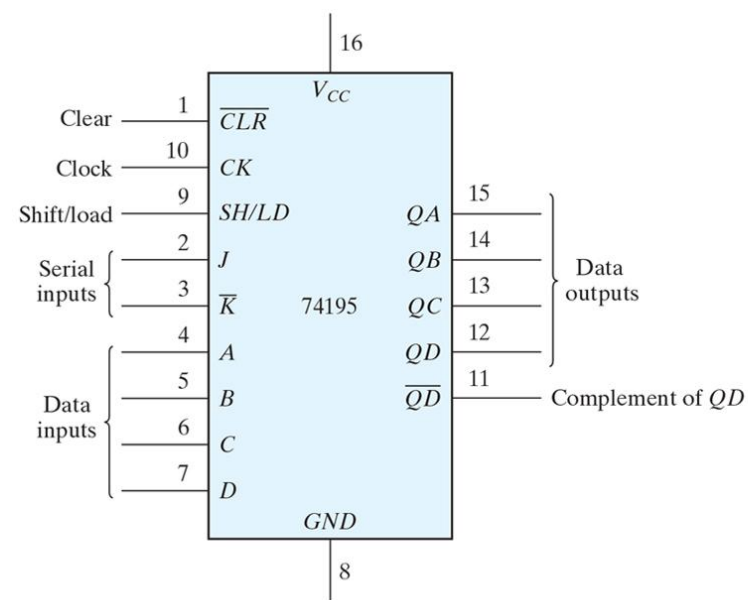
Figure 9.15
IC type 74161 binary counter with parallel load.



Function table

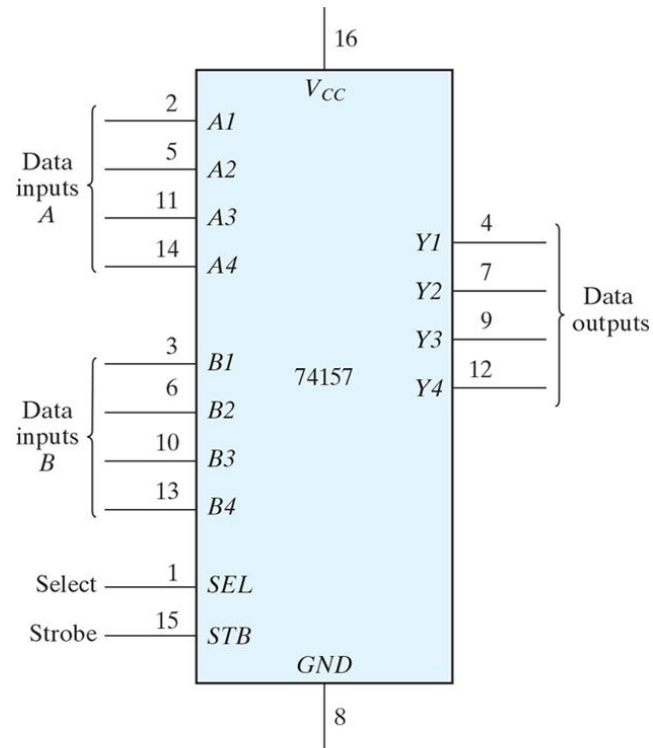
Clear	Clock	Load	Count	Function
0	X	X	X	Clear outputs to 0
1	↑	0	X	Load input data
1	↑	1	1	Count to next binary value
1	↑	1	0	No change in output

Figure 9.16
IC type 74195 shift register with parallel load.



Function table						
Clear	Shift/ load	Clock	J	$\overline{K}$	Serial input	Function
0	X	X	X	X	X	Asynchronous clear
1	X	0	X	X	X	No change in output
1	0	$\uparrow$	X	X	X	Load input data
1	1	$\uparrow$	0	0	0	Shift from QA toward QD , $QA = 0$
1	1	$\uparrow$	1	1	1	Shift from QA toward QD , $QA = 1$

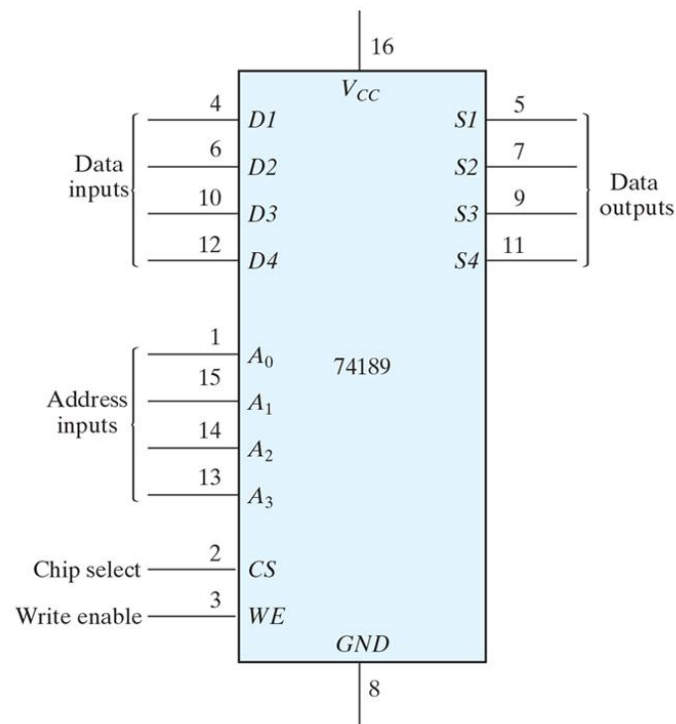
Figure 9.17
IC type 74157 quadruple 2 × 1 multiplexers.



Function table

Strobe	Select	Data outputs Y
1	X	All 0's
0	0	Select data inputs A
0	1	Select data inputs B

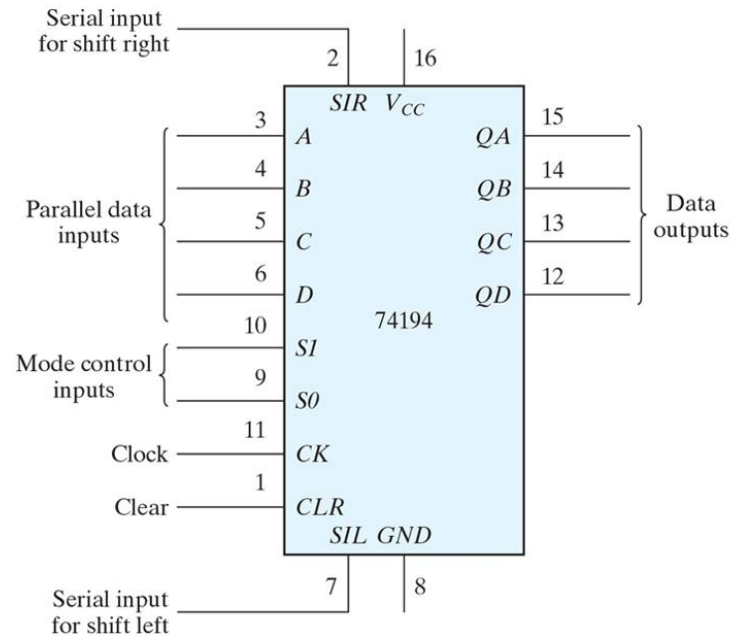
Figure 9.18
IC type 74189 16 × 4 RAM.



Function table

CS	WE	Operation	Data outputs
0	0	Write	High impedance
0	1	Read	Complement of selected word
1	X	Disable	High impedance

Figure 9.19
IC type 74194 bidirectional shift register with parallel load.



Function table

Clear	Clock	Mode		Function
		S_1	S_0	
0	X	X	X	Clear outputs to 0
1	$\uparrow$	0	0	No change in output
1	$\uparrow$	0	1	Shift right in the direction from QA to QD . SIR to QA
1	$\uparrow$	1	0	Shift left in the direction from QD to QA . SIL to QD
1	$\uparrow$	1	1	Parallel-load input data

Figure 9.20
Lamp handball logic diagram.

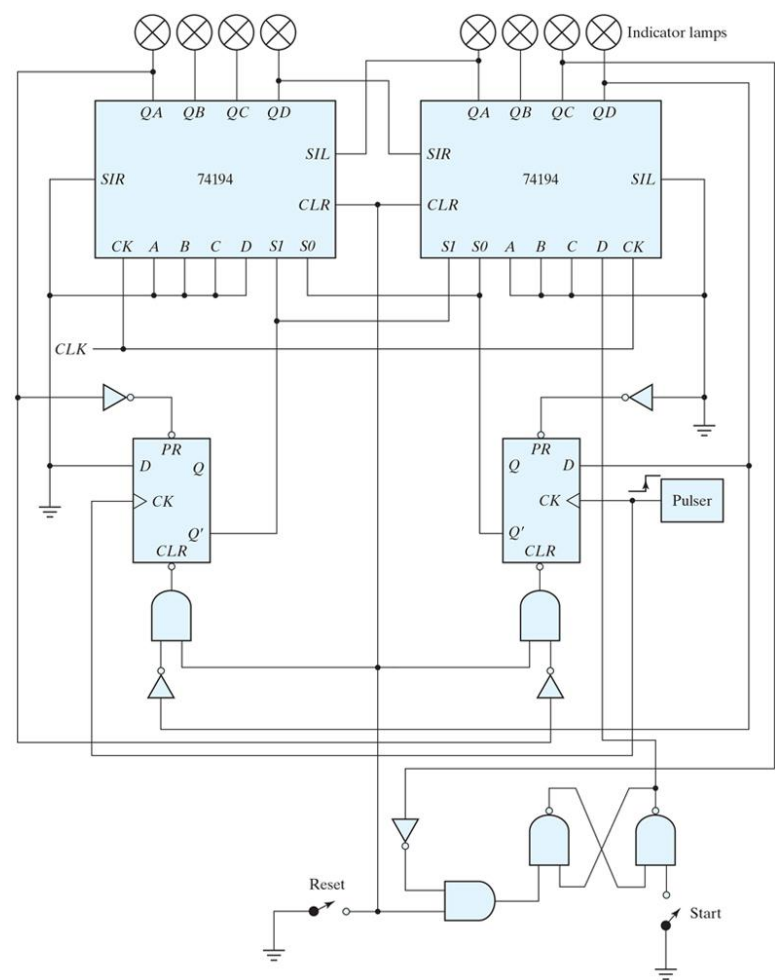


Figure 9.21
IC type 72555 timer connected as a clock-pulse generator.

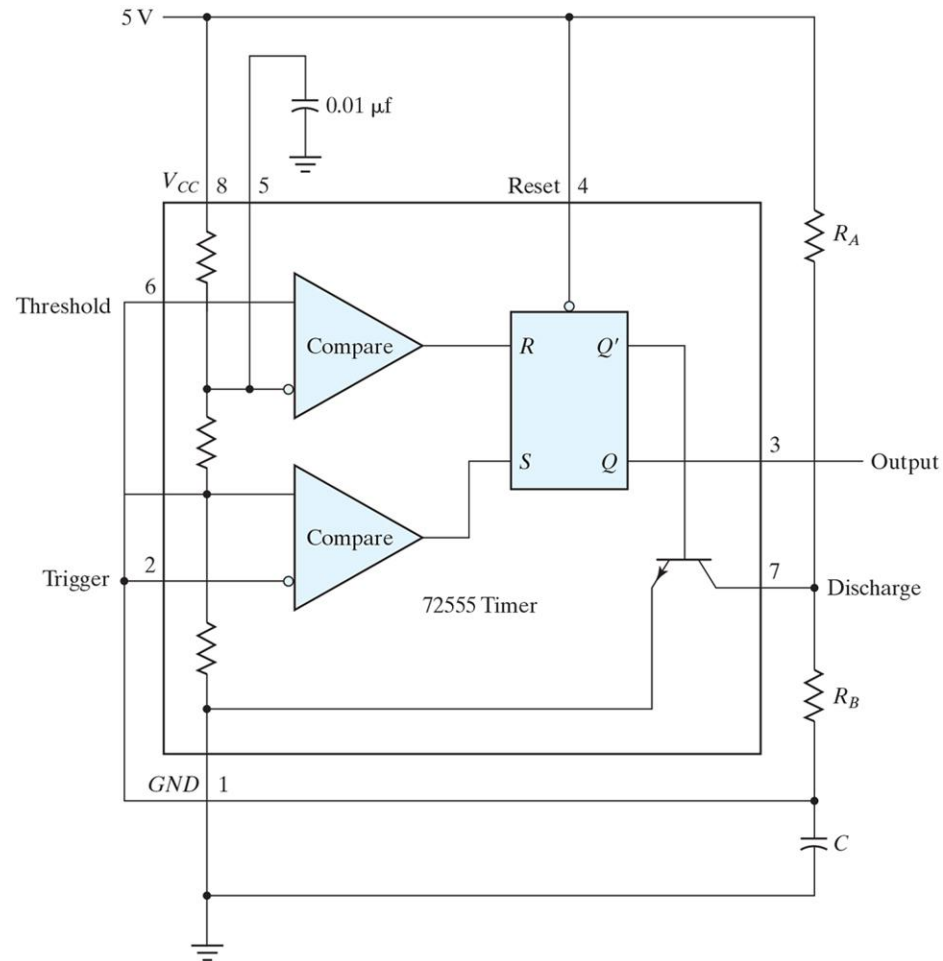


Figure 9.22
Output waveform for clock generator.

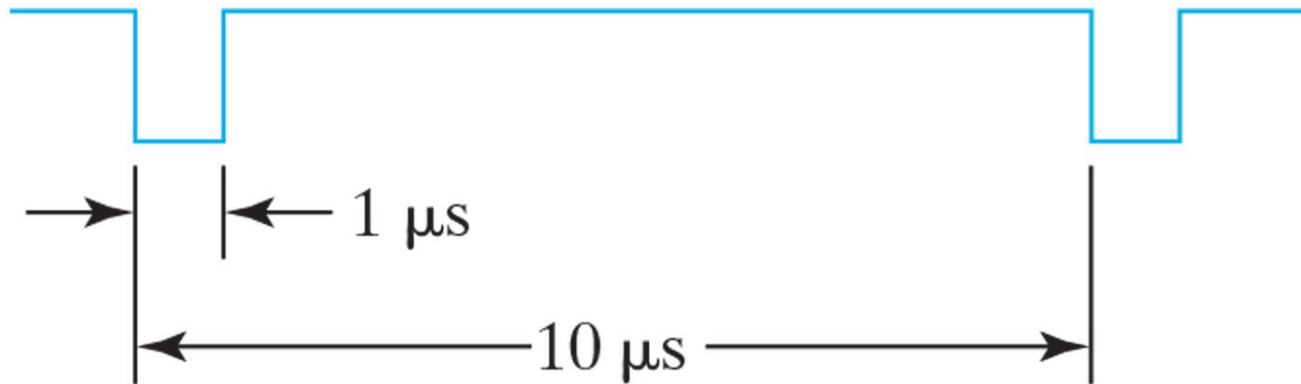


Figure 9.23
Block diagram of a parallel adder for Experiment 16.

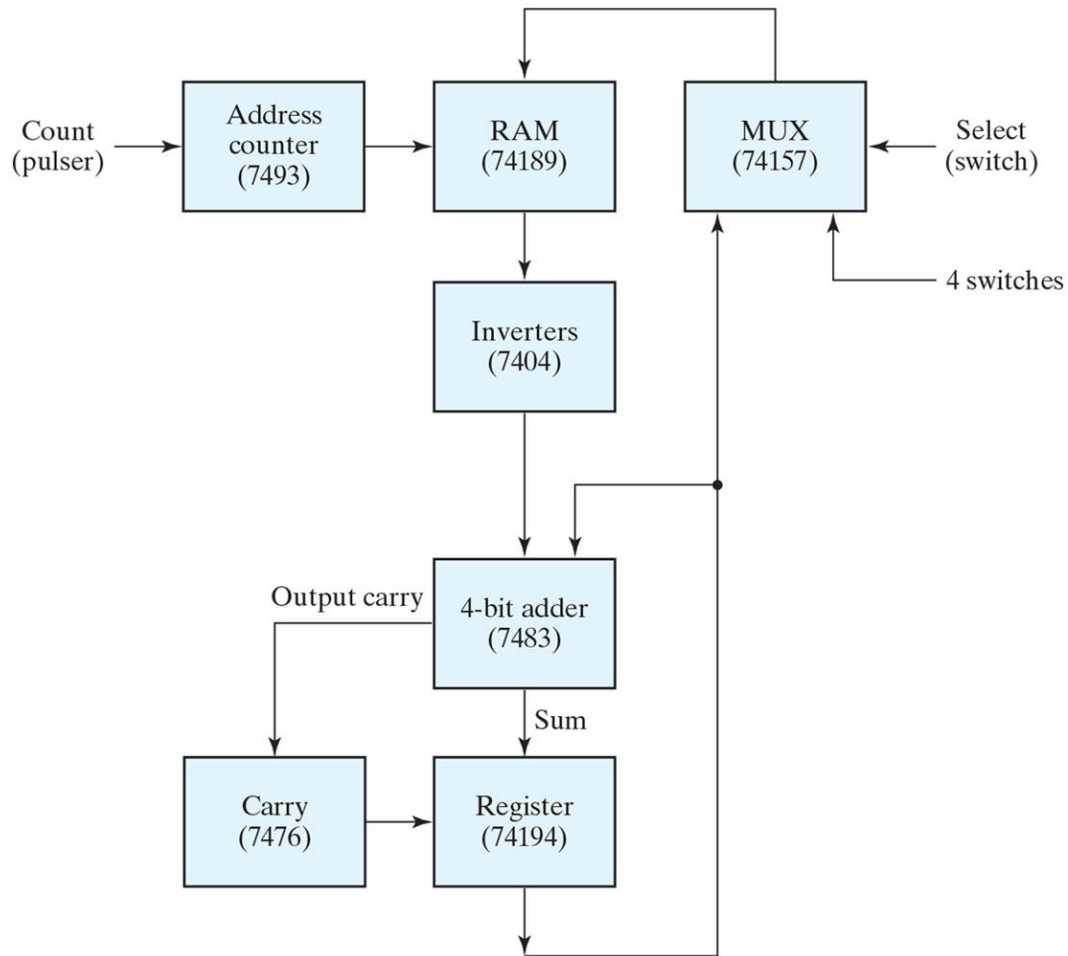
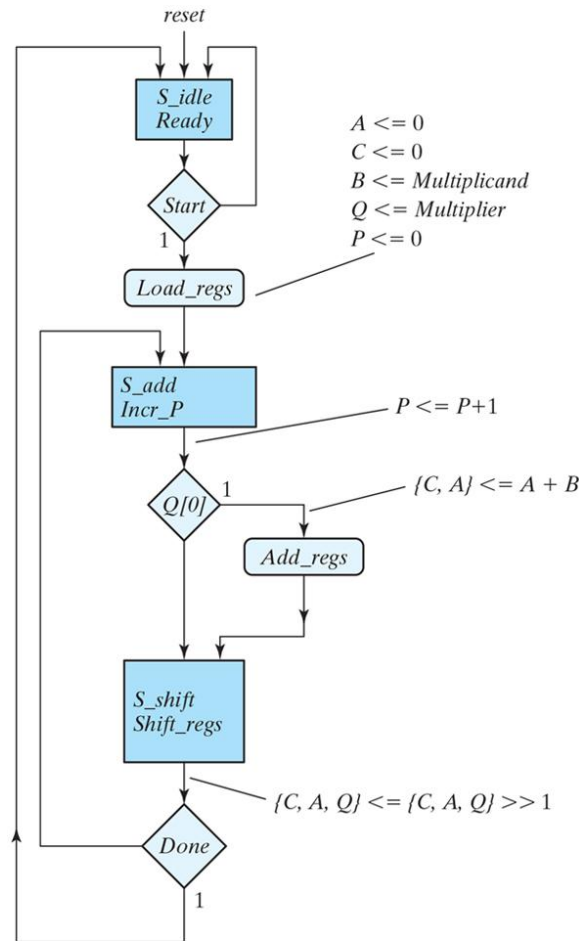
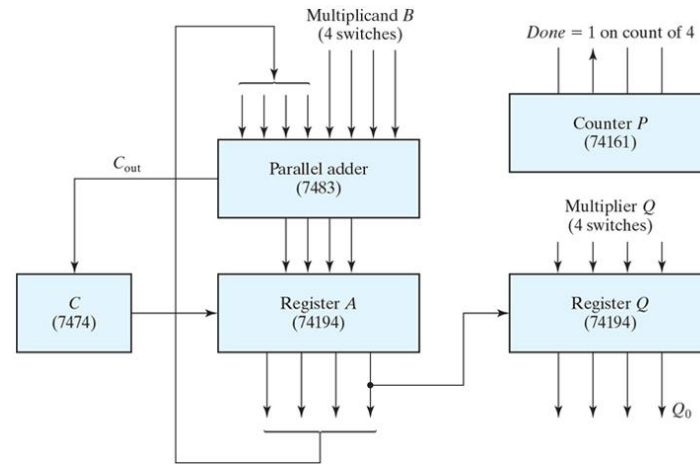


Figure 9.24
ASMD chart, block diagram of the datapath, control state diagram,
and register operations of the binary multiplier circuit.

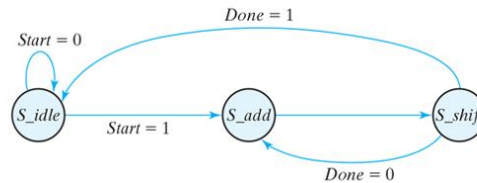


(a) ASMD chart

Figure 9.24



(b) Datapath block program



(c) Control state diagram

State Transition		Register Operations	Control signals
<u>From</u>	<u>To</u>		
S_idle		Initial state reached by reset action	
S_idle	S_add	$A <= 0, C <= 0, P <= 0$ $B <= Multipl\>icand, Q <= Multiplier$	$Load_regs$
S_add	S_shift	$P <= P + 1$ if $(Q[0])$ then $(A <= A + B, C <= C_{out})$	$Incr_P$ Add_regs
S_shift		shift right $\{CAQ\}, C <= 0$	$Shift_regs$

(d) Register operations

Figure 9.25
Pulser circuit for FPGA implementation of Experiment 1.

